



Client-use note. Prepared as prospect-facing GTM collateral from representative engineering datasets and cited references. Class-typical vessel values are not a substitute for owner-approved vessel data, project-specific metocean, certified RAOs, or final installation engineering.

DIGITALMODEL — ENGINEERING INTELLIGENCE

Deepwater Mudmat Installation Analysis

180 parametric cases: 2 vessels x 6 depths x 3 mudmats x 5 sea states

180 parametric cases analysed overnight

Generated 2026-04-15 04:04 UTC · demo_03

Methodology

This analysis screens deepwater mudmat installations across a parametric matrix of vessels, water depths, structure sizes, and sea states. Each case evaluates five installation phases per DNV-RP-H103 and DNV-ST-N001 methodology.

Phase 1: Lift-off

Hook load = $mass_{air} \times g \times DAF_{lift}$ (1.10). Checked against crane SWL at operating radius via interpolation of the crane capacity curve.

Phase 2: In-air Transit

Tilt check for 4-point lift with spreader beam. CoG offset / hook height gives tilt angle, checked against 5° limit. Symmetric mudmats always pass.

Phase 3: Splash Zone (Critical)

Slamming force: $F_{slam} = 0.5 \times \rho \times C_s \times A_p \times v_{rel}^2$, where $v_{rel} = v_{lowering} + v_{wave}$. Additional varying buoyancy and drag forces. $DAF_{splash} = 1.30$. This phase typically governs for large mudmats in higher sea states.

Phase 4: Lowering through Water Column

Cable tension = $W_{sub} + W_{cable}(depth) + snap\ load$. Snap load from dynamic cable response using vessel heave RAO. Checked against 85% of wire MBL. Becomes critical at extreme depths due to cable self-weight.

Phase 5: Landing

Bearing pressure = W_{sub} / A_{base} . Checked against 50 kPa soft clay bearing capacity. Rarely governs for typical mudmat proportions.

Go/No-Go Criteria

- GO:** All phases pass AND max utilisation < 0.85
- MARGINAL:** All phases pass AND max utilisation in [0.85, 1.00]
- NO-GO:** Any phase fails (utilisation > 1.00)

Applicable Codes & Standards

- DNV-RP-H103 (2011) — Modelling and Analysis of Marine Operations
- DNV-ST-N001 (2021) — Marine Operations and Marine Warranty
- DNV-RP-C205 (2010) — Environmental Conditions and Environmental Loads

Installation Screening Summary

All cases at $H_s = 2.0$ m reference sea state

VESSEL	STRUCTURE	DEPTH (M)	MAX UTIL.	GOVERNING PHASE	STATUS
Large CSV	Mudmat-S-50te	500	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	500	21.5%	Landing	GO

VESSEL	STRUCTURE	DEPTH (M)	MAX UTIL.	GOVERNING PHASE	STATUS
Large CSV	Mudmat-L-200te	500	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	1000	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	1000	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	1000	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	1500	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	1500	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	1500	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	2000	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	2000	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	2000	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	2500	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	2500	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	2500	20.6%	Splash zone	GO
Large CSV	Mudmat-S-50te	3000	24.7%	Landing	GO
Large CSV	Mudmat-M-100te	3000	21.5%	Landing	GO
Large CSV	Mudmat-L-200te	3000	20.6%	Splash zone	GO
Medium CSV	Mudmat-S-50te	500	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	500	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	500	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	1000	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	1000	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	1000	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	1500	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	1500	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	1500	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	2000	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	2000	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	2000	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	2500	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	2500	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	2500	71.5%	Splash zone	GO
Medium CSV	Mudmat-S-50te	3000	24.7%	Landing	GO
Medium CSV	Mudmat-M-100te	3000	34.8%	Splash zone	GO
Medium CSV	Mudmat-L-200te	3000	71.5%	Splash zone	GO

Instead of relying on forecasted Hs limits, the installation screening updates continuously with actual crane tip motions, hook loads, and dynamic amplification factors measured on your vessel.

Schedule a Technical Demo →

Assumptions & Limitations

- Crane SWL evaluated at maximum capacity radius (most favourable position)
- DAF lift-off = 1.10, DAF splash zone = 1.30 per DNV-ST-N001
- JONSWAP peak period approximation: $T_p = 4 * \sqrt[4]{H_s}$
- Slamming coefficient $C_s = 5.0$ for flat-bottom structures per DNV-RP-H103 Table 4-1
- Added mass coefficient $C_a = 1.0$, drag coefficient $C_d = 2.0$ for flat plate geometry
- Wire rope MBL safety factor = 0.85 (max tension / MBL)
- Bearing capacity limit = 50 kPa (soft clay, undrained)
- Vessel heave at crane tip uses simplified single-frequency RAO peak
- No current loads or wind loads on structure during lowering
- Rigging weight excluded from hook load calculations
- Seabed assumed flat — no slope corrections for landing
- Seawater density = 1025 kg/m³ throughout water column

Side-by-side vessel comparison at $H_s = 2.0$ m reference sea state.

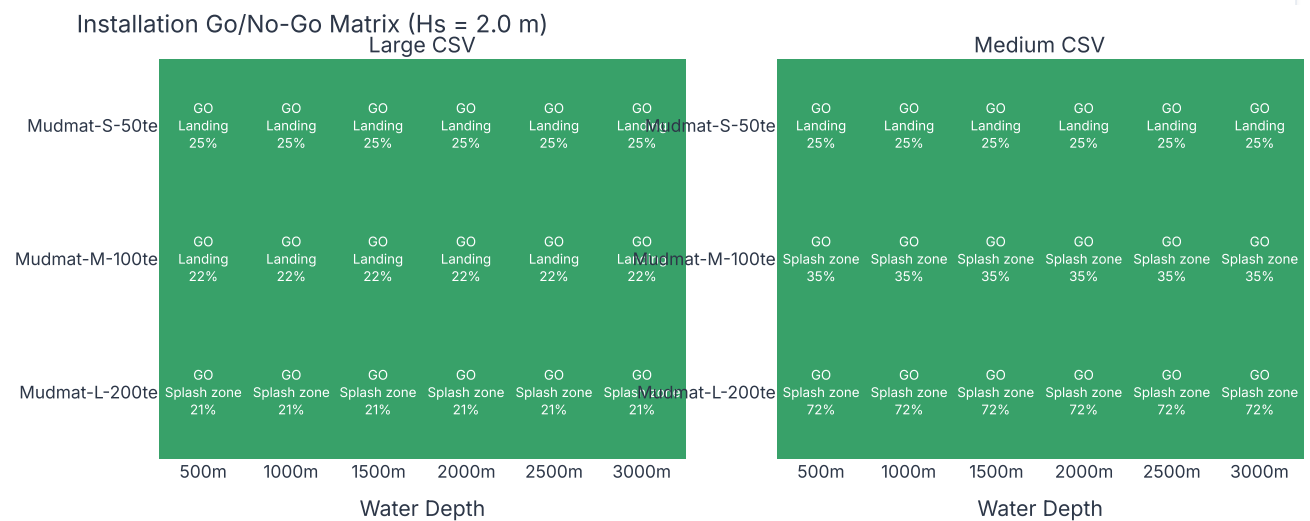


Chart 2: Crane Utilisation vs Water Depth

Max utilisation across all phases. Solid = Large CSV, dashed = Medium CSV.

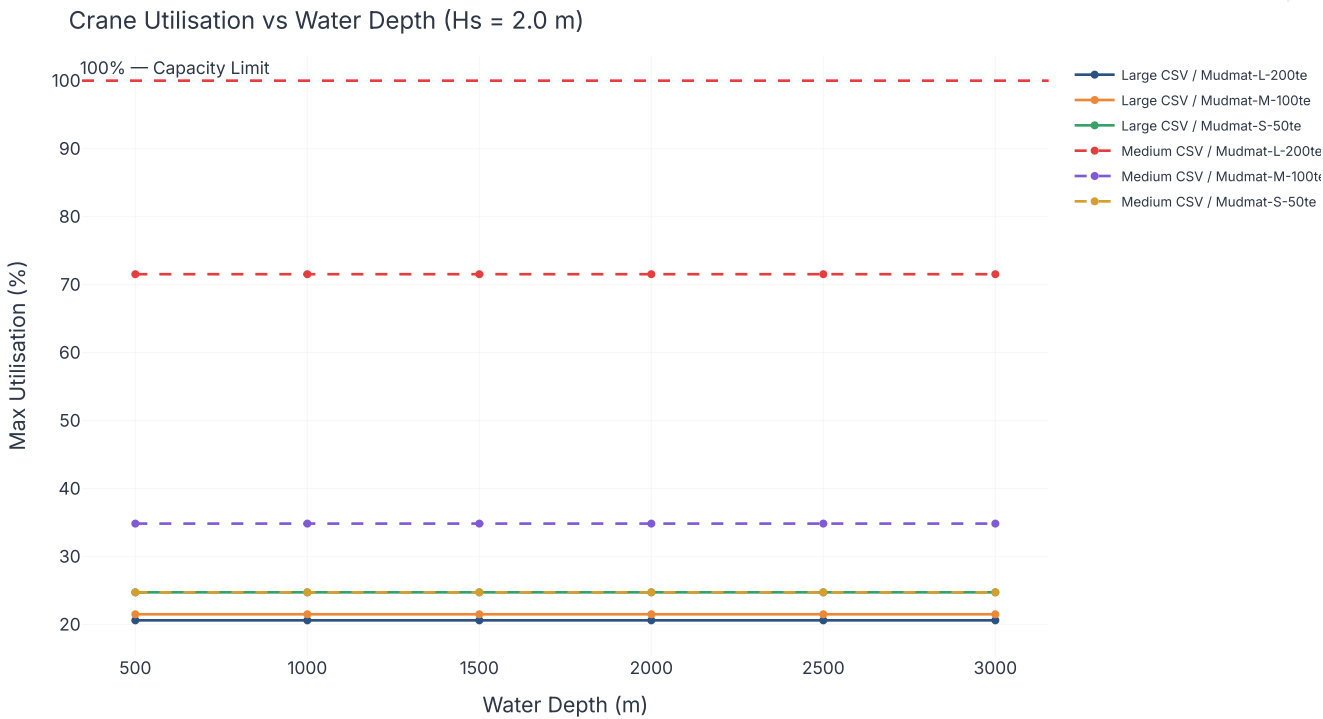


Chart 3: Phase Utilisation vs Water Depth — Splash & Lowering

Shows how splash zone dominates shallow cases while lowering governs at depth.

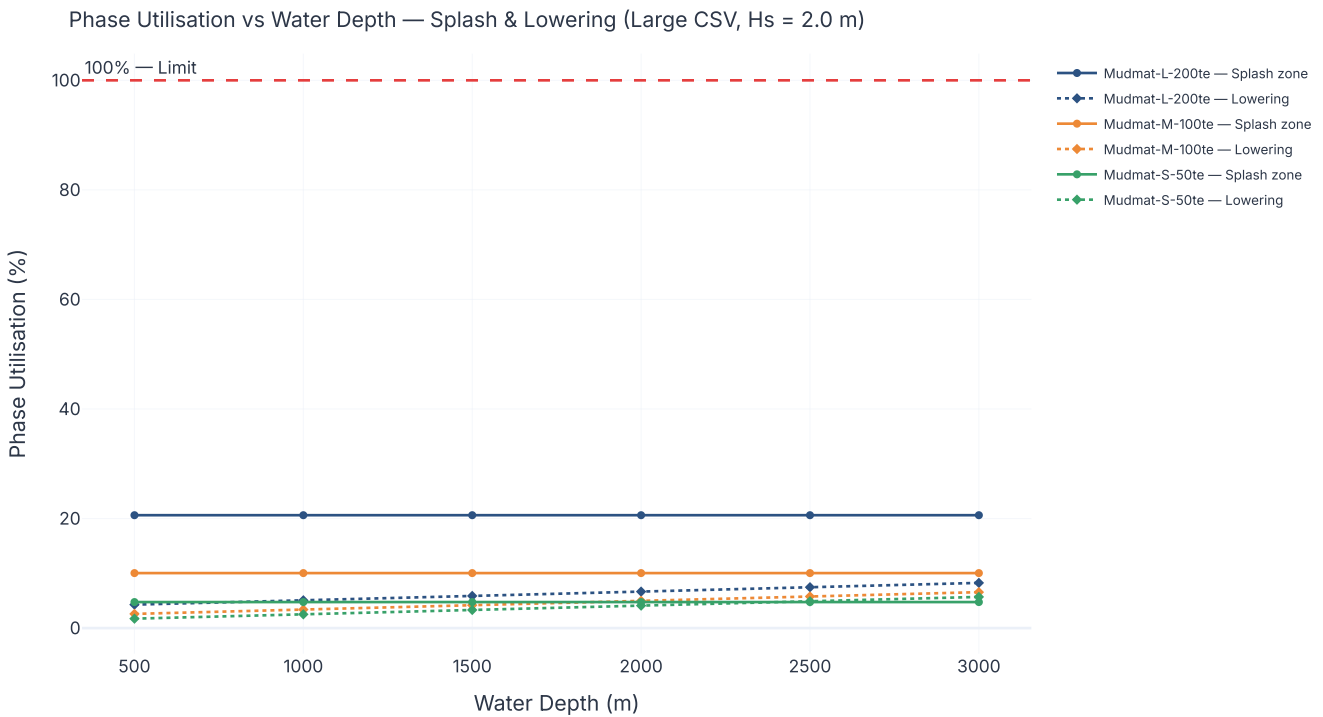


Chart 4: Operability Envelope — Max Hs vs Structure Weight

Maximum sea state where all depths pass. Defines weather window requirements.

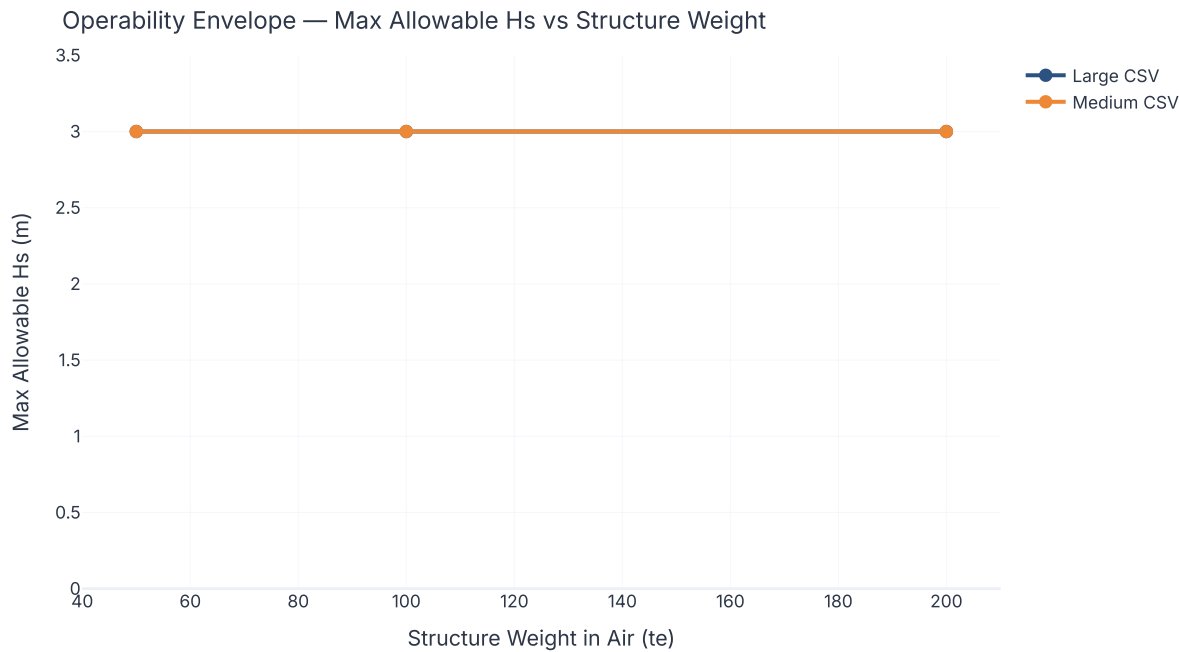
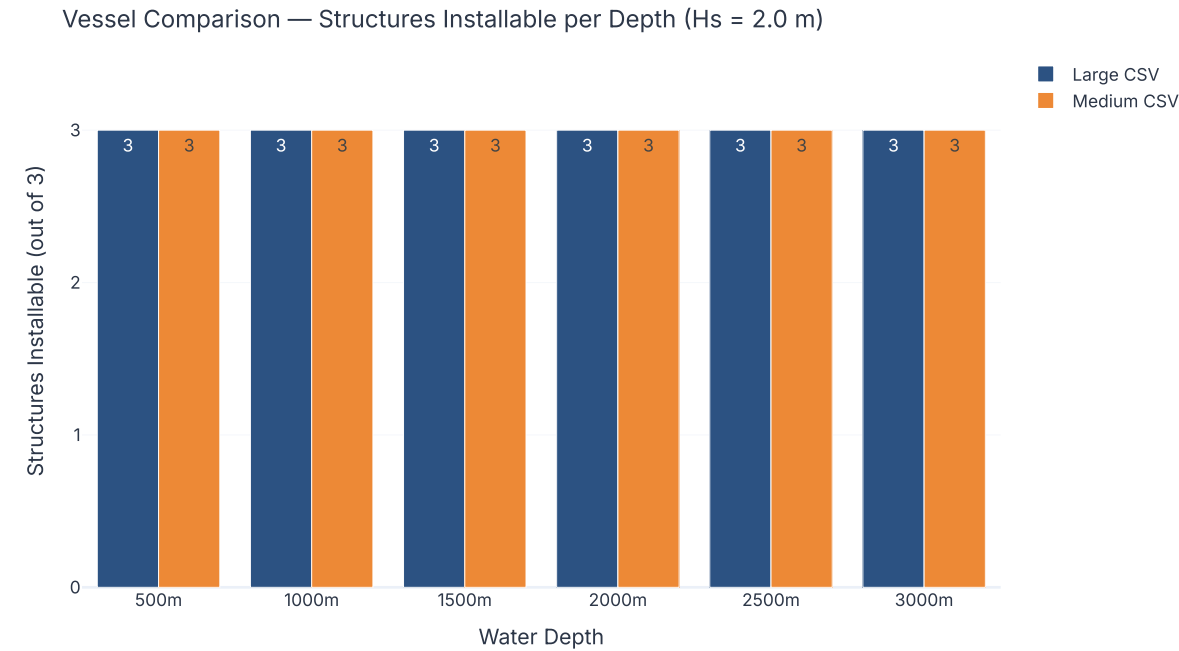


Chart 5: Vessel Head-to-Head — Structures Installable per Depth

Number of mudmat sizes (out of 3) each vessel can install at Hs = 2.0 m.



These results are preliminary engineering estimates generated by automated parametric analysis. All outputs require review by a qualified engineer before use in design or operational decisions.

References: DNV-RP-H103 (2011) — Modelling and Analysis of Marine Operations · DNV-ST-N001 (2021) — Marine Operations and Marine Warranty · DNV-RP-C205 (2010) — Environmental Conditions and Environmental Loads
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